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ORIGINAL ARTICLE

Analytic orbit theory with any arbitrary spherical harmonic as the dominant perturbation

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Abstract

In the gravitational potential of Earth, the oblateness term is the dominant perturbation, with its coefficient at least three orders of magnitude greater than that of any other zonal or tesseral spherical harmonic. Therefore, analytic orbit theories (or satellite theories) are developed using the Keplerian Hamiltonian as the unperturbed solution, oblateness term as the first-order and the remaining spherical harmonics as the second-order perturbations. These orbit theories are generally constructed by applying multiple near-identity canonical transformations to the perturbed Hamiltonian in conjunction with averaging to obtain a secular Hamiltonian, from which the short-period and long-period terms are removed. If the oblateness term is the only first-order perturbation, then the long-period terms appear only in the second- or higher-order terms of the single-averaged Hamiltonian, from which the short-period terms are removed. These second-order long-period terms are separated from the Hamiltonian by the first-order generating function using a second canonical transformation. This results in a secular Hamiltonian dependent only on the momenta. However, in the case of other gravitational bodies with more deformed shapes compared to Earth such as moons and asteroids, the oblateness coefficient may have the same order of magnitude as some of the higher spherical harmonic coefficients. If these higher harmonics are treated as the first-order perturbation along with the oblateness term, then the long-period terms appear in the first-order single-averaged Hamiltonian. These first-order long-period terms cannot be separated using the generating function in the conventional way. This problem occurs because the zeroth-order Hamiltonian, i.e., the Keplerian part, is degenerate in the angular momentum. In this paper, a new approach to the long-period transformation is proposed to resolve this issue and obtain a fully analytic orbit theory when for the perturbing gravitational body, any arbitrary zonal or tesseral harmonic is the dominant perturbation. The proposed theory is closed form in the eccentricity as well. It is applied to predict the motion of artificial satellites for the two test cases: a lunar orbiter and a satellite of 433 Eros asteroid. The prediction accuracy is validated against the numerical propagation using a force model with 6×6 gravity field.

This article is part of the topical collection on 50 years of Celestial Mechanics and Dynamical Astronomy
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1 Introduction

The beginning of the space age generated numerous analytic orbit theories for estimating the orbits of space objects (Mahajan 2018). Brouwer (1959) used von Zeipel's method to obtain a perturbation solution to the artificial satellite problem including the zonal harmonics from J_2 to J_5 of the geopotential. His theory includes the secular terms up to second order and short-period and long-period terms up to first order. In contrast, Vinti (1959) obtained a closed-form solution using the spheroidal coordinates for the motion of an object about an oblate spheroid. Garfinkel (1959) also obtained a closed-form solution of a modified geopotential that included a portion of the oblateness potential. Kozai (1959) developed a similar theory to that of Brouwer by separating the secular, short-period, and long-period terms from the disturbing potential and then computed the variations of the orbital elements using Lagrange's planetary equations. Of these, Brouwer's theory with the addition of the atmospheric drag effects emerged as the basis for orbit prediction in the USA and maintaining the space object catalog.

von Zeipel's method of performing canonical transformations suffer from a drawback of producing mixed-term generating functions. First Hori (1966) and then Deprit (1969) proposed the use of near-identity transformations for constructing the canonical transformations, which avoids the limitations of von Zeipel's method. Subsequently, many special canonical transformations (Deprit 1981a; Alfriend and Coffey 1984; Coffey and Alfriend 1981) were proposed to transform the Hamiltonian in a simplified form suitable for further normalization. Using these transformations, the orbit theories for the artificial satellites of Earth have been constructed up to third and higher orders with the oblateness term considered as the first-order perturbation (Deprit 1981b; Coffey and Deprit 1982; Kudryavtsev 1995; Kinoshita 1978).

In the geopotential, the oblateness coefficient (J_2) is of $\mathscr{O}(10^{-3})$ and all the other coefficients have magnitudes of $\mathscr{O}(J_2^2)$ or smaller. Therefore, the perturbation solutions are generally developed in powers of J_2 with the oblateness term as the first-order perturbation and the rest of the harmonics treated at the second order. When using the canonical perturbation methods, the short-period averaging of the original Hamiltonian produces long-period terms, which are dependent on the argument of periapsis, in the new averaged Hamiltonian. These long-period terms occur for all the spherical harmonics included in the original Hamiltonian except for the oblateness term. As a result, these long-period terms first appear at the second order and because the Keplerian Hamiltonian is degenerate in the angular momentum, these second-order terms are eliminated from the single-averaged Hamiltonian using the first-order term of the long-period generating function.

However, a problem occurs when any of the higher harmonic coefficients are of $\mathscr{O}(J_2)$ and must be treated as first-order perturbations. Then, the corresponding long-period terms appear at the first order in the single-averaged Hamiltonian and they cannot be separated from it using a long-period transformation in the conventional way. This is due to the fact that the zeroth-order (Keplerian) Hamiltonian is degenerate in the angular momentum, and thus, no partials of the generating function with respect to the argument of periapsis are available to eliminate the first-order long-period terms. As a result, the long-period transformation does not result in a Hamiltonian that is a function of only the momenta; consequently, one does

not obtain a set of mean elements having only secular rates. In this paper, a novel approach is developed for addressing this problem and obtaining a secular Hamiltonian resulting in coordinates that are completely secular or in other words vary linearly with time. The proposed approach can be implemented using a two-step averaging process. The first step is the standard averaging process for the short-period terms for an arbitrary zonal or tesseral harmonic using the Keplerian Hamiltonian as the zeroth-order Hamiltonian, carried out to the desired order. In this work, the short-period averaging is, in fact, performed separately for the zonal and tesseral harmonics. The result is a new Hamiltonian that does not depend on the true or mean anomaly. Then, in the second step, the secular portion of this Hamiltonian with contributions from all the spherical harmonics considered is chosen as the new zeroth-order Hamiltonian. The remaining long-period part of the Hamiltonian is treated as the perturbation. This new zeroth-order non-Keplerian Hamiltonian is non-degenerate in the angular momentum, and the long-period averaging process can now be used successfully to separate the long-period terms.

The following sections present the complete mathematical development of a first-order orbit theory using a complete gravitational potential with all the zonal, sectorial, and tesseral harmonics considered as the dominant perturbations. The short-period terms due to the sectorial and tesseral harmonics are eliminated first from the Hamiltonian in Sect. 3. This is followed by another transformation for separating the zonal short-period terms due to the zonal harmonics. In Sect. 4, the long-period terms are separated from the averaged Hamiltonian obtained in the previous section in order to completely normalize the original Hamiltonian. The formulae for the averaged Hamiltonians and the corresponding generating functions are derived in a generalized form, which makes them valid for any arbitrary spherical harmonic. Section 5 applies the proposed theory to the motion of a lunar orbiter and a satellite of 433 Eros asteroid, followed by conclusions in Sect. 6.

2 Preliminaries

The Hamiltonian for a perturbed Keplerian orbit may be expressed in the form of a power series as

$$\mathscr{H} = \mathscr{H}_0 + \sum_{n=1} \frac{\epsilon^n}{n!} \mathscr{H}_n, \tag{1}$$

where $\mathscr{H}_0$ is the zeroth-order Keplerian part, $\mathscr{H}_n$ is the perturbation term of order n, and ϵ is the formal parameter used to separate the perturbation terms according to their respective order. ϵ is set to unity in the final results. The methodology for the Delaunay normalization of the above Hamiltonian using the Lie-transform-based methods (Deprit 1969; Kamel 1969) rather than the von Zeipel's method originally used by Brouwer (1959) is given here for completion. The normalization is carried out by using a near-identity canonical transformation from the original canonical variables $(\mathbf{x}, \mathbf{X})$ to the new canonical variables $(\mathbf{y}, \mathbf{Y})$. The new Hamiltonian $\mathscr{K}$ dependent on $(\mathbf{y}, \mathbf{Y})$, sometimes referred to as Kamiltonian, may also be expressed as a power series as shown below:

$$\mathscr{K}(\mathbf{y}, \mathbf{Y}) = \mathscr{K}_0(\mathbf{y}, \mathbf{Y}) + \sum_{n=1} \left(\frac{\epsilon^n}{n!} \right) \mathscr{K}_n(\mathbf{y}, \mathbf{Y}). \tag{2}$$

It is noted that the lower-case and upper-case canonical variables represent the coordinates and momenta, respectively, with boldface indicating vector quantities. The homological equa-

tions relate the old and new Hamiltonian terms and are given below up to second order (Deprit 1969):

$$\mathscr{K}_0 = \mathscr{H}_0, \tag{3}$$

$$\mathscr{K}_1 = (\mathscr{H}_0, \mathscr{W}_1) + \mathscr{H}_1 - \frac{\partial \mathscr{W}_1}{\partial t}, \tag{4}$$

$$\mathscr{K}_2 = \mathscr{H}_2 + (\mathscr{H}_1 + \mathscr{K}_1, \mathscr{W}_1) + (\mathcal{H}_0, \mathcal{W}_2) - \frac{\partial \mathscr{W}_2}{\partial t}, \tag{5}$$

where $(-,-)$ represents Poisson bracket, which is defined with respect to the canonical variables $(\mathbf{y}, \mathbf{Y})$ as

$$(A, B) = \frac{\partial A}{\partial \mathbf{y}} \left(\frac{\partial B}{\partial \mathbf{Y}}\right)^{\mathrm{T}} - \frac{\partial A}{\partial \mathbf{Y}} \left(\frac{\partial B}{\partial \mathbf{y}}\right)^{\mathrm{T}}.$$

The superscript T denotes the transpose operator. The old and new variables are related through the following near-identity canonical transformations:

$$\mathbf{x} = \mathbf{y} + \epsilon(\mathbf{y}, \mathscr{W}_1) + \frac{\epsilon^2}{2!}\left[(\mathbf{y}, (\mathbf{y}, \mathscr{W}_1)) + (\mathbf{y}, \mathscr{W}_2)\right] + \mathscr{O}(\epsilon^3), \tag{6}$$

$$\mathbf{X} = \mathbf{Y} + \epsilon(\mathbf{Y}, \mathscr{W}_1) + \frac{\epsilon^2}{2!}\left[(\mathbf{Y}, (\mathbf{Y}, \mathscr{W}_1)) + (\mathbf{Y}, \mathscr{W}_2)\right] + \mathscr{O}(\epsilon^3). \tag{7}$$

The new variables can also be expressed in terms of the old variables using the inverse transformation equations

$$\mathbf{y} = \mathbf{x} - \epsilon(\mathbf{x}, \mathscr{W}_1) + \frac{\epsilon^2}{2!}\left[(\mathbf{x}, (\mathbf{x}, \mathscr{W}_1)) - (\mathbf{x}, \mathscr{W}_2)\right] + \mathscr{O}(\epsilon^3), \tag{8}$$

$$\mathbf{Y} = \mathbf{X} - \epsilon(\mathbf{X}, \mathscr{W}_1) + \frac{\epsilon^2}{2!}\left[(\mathbf{X}, (\mathbf{X}, \mathscr{W}_1)) - (\mathbf{X}, \mathscr{W}_2)\right] + \mathscr{O}(\epsilon^3). \tag{9}$$

It is noted that the generating function, $\mathscr{W}$, is a function of $(\mathbf{y},\mathbf{Y})$ and $(\mathbf{x},\mathbf{X})$ in Eqs. (6–7) and (8–9), respectively.

In order to perform the Delaunay normalization of the gravitational potential using the canonical transformations, it needs to be first expressed in terms of the canonical variables. The Delaunay elements are used in this work, which are defined as

$$\begin{aligned} l &= M, & L &= \sqrt{\mu a}, \\ g &= \omega, & G &= L\eta, \\ h &= \Omega, & H &= G\cos i, \end{aligned} \tag{10}$$

where $\eta = \sqrt{1-e^2}$ and the conventional notation is used for the classical orbital elements. The gravitational potential, in spherical coordinates with the zonal, sectorial, and tesseral spherical harmonics included, is (Kaula 1966)

$$V = -\frac{\mu}{r}\left[1 - \sum_{n=2}^{\infty} J_n \left(\frac{R_e}{r}\right)^n P_n(\sin\phi) + \sum_{n=2}^{\infty}\sum_{m=1}^{n}\left(\frac{R_e}{r}\right)^n P_{nm}(\sin\phi)\left(C_{nm}\cos m\lambda + S_{nm}\sin m\lambda\right)\right], \tag{11}$$

where μ is the gravitational parameter, R_e is the representative radius for the gravitational body under consideration, (r, ϕ, λ) are the radius, latitude, and longitude of the artificial satellite, and P_n and P_{nm} are the Legendre functions of the first kind and associated Legendre functions, respectively. The spherical harmonic degree and order are represented by n and m, respectively. Using a mix of the classical and Delaunay elements, the perturbation part of the above potential for an arbitrary spherical harmonic of degree n and order m can be written as (Kaula 1966)

$$V_{nm} = \frac{\mu R_e^n}{r^{n+1}} \sum_{p=0}^{p=n} F_{nmp}(i)\left[CS_1\cos(\Psi) + CS_2\sin(\Psi)\right], \tag{12}$$

where

$$\Psi \equiv (n-2p)(f+g) + m(h-\theta),$$

$$CS_1 = \begin{cases} C_{nm} & \text{if } n-m \text{ is even} \\ -S_{nm} & \text{if } n-m \text{ is odd,} \end{cases}$$

$$CS_2 = \begin{cases} S_{nm} & \text{if } n-m \text{ is even} \\ C_{nm} & \text{if } n-m \text{ is odd,} \end{cases}$$

and θ represents the sidereal time, a measure of the central body's rotation. The inclination function $F_{nmp}(i)$ is given by (Yan et al. 2014; Emeljanov and Kanter 1989)

$$F_{nmp}(i) = (-1)^{E\left(\frac{n-m+1}{2}\right)} \frac{(n+m)!}{2^n p!(n-p)!} \sum_{j=\max(n-2p-m,\,0)}^{j=\min(n-m,\,2n-2p)} (-1)^j \times \binom{2n-2p}{j}\binom{2p}{n-m-j}\cos^{2n-s}\left(\frac{i}{2}\right)\sin^s\left(\frac{i}{2}\right), \tag{13}$$

where $s = m - n + 2p + 2j$ and $E()$ represents the integer part function.

For Earth, the oblateness coefficient $J_2(= -C_{20}) \approx 0.00108$, $J_n(= -C_{n0})$ is of $\mathscr{O}(J_2^2)$ for $n > 2$, and C_{nm}/S_{nm} is of $\mathscr{O}(J_2^2)$ for $n \geq 2$, $m = 1, \ldots, n$. However, this may not be true for the gravitational bodies with more deformed shapes. To address this type of scenarios in a general manner, all the spherical harmonic terms are considered to be of the same order of magnitude in this paper and are collected in the first-order Hamiltonian term $\mathscr{H}_1$ of Eq. 1 to obtain the complete Hamiltonian as

$$\mathscr{H} = \mathscr{H}_0 + \epsilon\mathscr{H}_1, \tag{14}$$

where

$$\mathscr{H}_0 = -\frac{\mu^2}{2L^2},$$
$$\mathscr{H}_1 = -\sum_{n=2}^{\infty}\sum_{m=0}^{n} V_{nm}.$$

It is noted that the minus sign in $\mathscr{H}_0$ and $\mathscr{H}_1$ is due to the adopted convention of considering the Delaunay angle elements as the first three coordinates and the action elements as the last three.

3 Short-period averaging

For a first-order theory, there is no coupling among any of the zonal, sectorial, and tesseral harmonics. As a result, the complete Delaunay normalization of the Hamiltonian $\mathscr{H}$ as given in Eq. (14) can be performed using the three separate transformations. First, the short-period terms due to zonals and tesserals are separated from $\mathscr{H}$ using two transformations, and then, the long-period terms are separated using the long-period transformation to obtain a completely secular Hamiltonian. In this section, the two transformations for separating all the short-period terms from $\mathscr{H}$ are derived.

Tesseral short-period averaging The short-period effects due to the sectorial and tesseral harmonics are first eliminated from $\mathscr{H}$ by constructing a new Hamiltonian $\mathscr{K}$, which consists of all the zonal harmonic terms in addition to the Keplerian part. It is assumed in the present work that no resonant long-period or secular effects exist due to any of the sectorial or tesseral harmonics. The homological equations up to first order given in Eqs. (3–4), relating the old and the new Hamiltonians, are

$$\mathscr{K}_0 = \mathscr{H}_0, \tag{15}$$
$$\mathscr{K}_1 = (\mathscr{H}_0, \mathscr{W}_1) + \mathscr{H}_1 - \frac{\partial \mathscr{W}_1}{\partial t}. \tag{16}$$

In the preceding equations, $\mathscr{K}_1$ is chosen to consist of all the zonal harmonics (for which $m = 0$) present in $\mathscr{H}_1$ from Eq. (14) as shown below:

$$\mathscr{K}_1 = \sum_{n=2}^{\infty} \frac{\mu R_e^n}{r^{n+1}} \sum_{p=0}^{p=n} F_{n0p}(i) J_n A, \tag{17}$$

where

$$A = \begin{cases} \cos((n-2p)(f+g)) & \text{if } n \text{ is even} \\ \sin((n-2p)(f+g)) & \text{if } n \text{ is odd,} \end{cases} \tag{18}$$

and $J_n = -C_{n0}$. By substituting the preceding expression for $\mathscr{K}_1$ in the first-order homological equation in Eq. (16), the following partial differential equation (PDE) is obtained:

$$-\tilde{n}\frac{\partial \mathscr{W}_1}{\partial l} - w_E\frac{\partial \mathscr{W}_1}{\partial \theta} = \sum_{n=2}^{\infty}\sum_{m=1}^{n} V_{nm}, \tag{19}$$

where

$$\tilde{n} = \frac{\partial \mathscr{H}_0}{\partial L} = \frac{\mu^2}{L^3},$$

$$\frac{\partial}{\partial \theta} = \frac{1}{w_E}\frac{\partial}{\partial t},$$

and w_E is the angular velocity of the central body. This PDE must be solved in order to compute the first-order short-period generating function, $\mathscr{W}_1$, for the tesseral and sectorial harmonics. A closed-form solution of the above PDE is not known to exist (Segerman and Coffey 2000). However, an exact solution, expressed in terms of two quadratures, is obtained by Mahajan et al. (2018) by using the method of characteristics. This solution, as follows, is used in the present work:

$$\begin{aligned}\mathscr{W}_1 = -\sum_{n=2}^{\infty}\frac{\mu R_e^n}{\tilde{n}\, a^{n+1}\eta^{2n-1}}\sum_{p=0}^{p=n} F_{nmp}(i)\left[CS_1\left\{\cos(\beta)I_1^{nmp} - \sin(\beta)I_2^{nmp}\right\}\right.\\ \left.+CS_2\left\{\cos(\beta)I_2^{nmp} + \sin(\beta)I_1^{nmp}\right\}\right] + F(\theta - \delta\, l, a, e, i, h, g),\end{aligned} \tag{20}$$

where

$$\beta \equiv (n-2p)g + m(h - \theta + \delta\, l), \tag{21}$$

$$I_1^{nmp} \equiv \int \cos((n-2p)f - m\delta\, l)(1 + e\cos f)^{n-1}\,\mathrm{d}f, \tag{22}$$

$$I_2^{nmp} \equiv \int \sin((n-2p)f - m\delta\, l)(1 + e\cos f)^{n-1}\,\mathrm{d}f, \tag{23}$$

and δ is defined as the ratio of w_E and $\tilde{n}$. Using $\mathscr{W}_1$, the first-order short-period variations due to sectorial and tesseral harmonics can be computed by evaluating the Poisson brackets of the generating function with the canonical elements as shown in Eqs. (6–9). The results for the first-order short-period variations of the Delaunay elements due to the sectorial and tesseral harmonics are

$$\Delta l = (l, \mathscr{W}_1) = \left(\frac{2L}{\mu}\right)\frac{\partial \mathscr{W}_1}{\partial a} + \left(\frac{\eta^2}{eL}\right)\frac{\partial \mathscr{W}_1}{\partial e} + \left(\frac{-G}{L^2}\right)\frac{\partial \mathscr{W}_1}{\partial \eta}, \tag{24}$$

$$\Delta g = (g, \mathscr{W}_1) = \left(\frac{-\eta^2}{eG}\right)\frac{\partial \mathscr{W}_1}{\partial e} + \left(\frac{1}{L}\right)\frac{\partial \mathscr{W}_1}{\partial \eta} + \left(\frac{\cos i}{G\sin i}\right)\frac{\partial \mathscr{W}_1}{\partial i}, \tag{25}$$

$$\Delta h = (h, \mathscr{W}_1) = \left(\frac{-1}{G\sin i}\right)\frac{\partial \mathscr{W}_1}{\partial i}, \tag{26}$$

$$\Delta L = (L, \mathscr{W}_1) = -\frac{\partial \mathscr{W}_1}{\partial l} - \left(\frac{\mathrm{d}f}{\mathrm{d}l}\right)\frac{\partial \mathscr{W}_1}{\partial f}, \tag{27}$$

$$\Delta G = (G, \mathscr{W}_1) = -\frac{\partial \mathscr{W}_1}{\partial g}, \tag{28}$$

$$\Delta H = (H, \mathscr{W}_1) = -\frac{\partial \mathscr{W}_1}{\partial h}, \tag{29}$$

where Δ on the left hand side of the above equations indicates that these are short-period variations that must be added to the corresponding elements, whose time variations are determined by $\mathscr{K}$, in order to recover the osculating elements. The following partials of the two quadratures, I_1^{nmp} and I_2^{nmp}, are needed to compute the partials of $\mathscr{W}_1$ in the preceding equations:

$$\frac{\partial I_1^{nmp}}{\partial a} = m\frac{\mathrm{d}\delta}{\mathrm{d}a}\int l\,\sin((n-2p)f - m\delta l)(1+e\cos f)^{n-1}\,\mathrm{d}f, \tag{30}$$

$$\frac{\partial I_2^{nmp}}{\partial a} = -m\frac{\mathrm{d}\delta}{\mathrm{d}a}\int l\,\cos((n-2p)f - m\delta l)(1+e\cos f)^{n-1}\,\mathrm{d}f, \tag{31}$$

$$\begin{aligned}\frac{\partial I_1^{nmp}}{\partial e} =& \frac{1}{\eta^2}\int\Bigg(\frac{-3e(n-1)}{2}\cos(\alpha(f)) + \frac{(-(n-1)+4p)}{2}\cos(\alpha(f)-f)\\ &+\frac{((3n+1)-4p)}{2}\cos(\alpha(f)+f) + \frac{e(2p+1)}{4}\cos(\alpha(f)-2f)\\ &+\frac{e(2n-2p+1)}{4}\cos(\alpha(f)+2f)\Bigg)(1+e\cos f)^{n-1}\,\mathrm{d}f,\end{aligned} \tag{32}$$

$$\begin{aligned}\frac{\partial I_2^{nmp}}{\partial e} =& \frac{1}{\eta^2}\int\Bigg(\frac{-3e(n-1)}{2}\sin(\alpha(f)) + \frac{(-(n-1)+4p)}{2}\sin(\alpha(f)-f)\\ &+\frac{((3n+1)-4p)}{2}\sin(\alpha(f)+f) + \frac{e(2p+1)}{4}\sin(\alpha(f)-2f)\\ &+\frac{e(2n-2p+1)}{4}\sin(\alpha(f)+2f)\Bigg)(1+e\cos f)^{n-1}\,\mathrm{d}f,\end{aligned} \tag{33}$$

where

$$\alpha(f) \equiv (n-2p)f - m\delta l, \tag{34}$$

$$\frac{\partial I_1^{nmp}}{\partial f} = \cos((n-2p)f - m\delta l)(1+e\cos f)^{n-1}, \tag{35}$$

$$\frac{\partial I_2^{nmp}}{\partial f} = \sin((n-2p)f - m\delta l)(1+e\cos f)^{n-1}. \tag{36}$$

The integrals in the preceding equations can be evaluated using series expansions in terms of either the eccentricity or the ratio of $\tilde{n}$ and w_E. They can also be computed very accurately using the numerical quadrature methods (Mahajan et al. 2018).

Zonal short-period averaging The Hamiltonian $\mathscr{K}$, obtained in the preceding subsection, consists of only the zonal harmonic perturbations in addition to the two-body term. If the secular or long-period effects due to resonant tesseral harmonics are known, they can also be included in $\mathscr{K}$. However, in the present work these effects are ignored. For the zonal short-period transformation, it is noted that $\mathscr{K}$ is not a function of time. Therefore, the homological equations relating $\mathscr{K}$ to the new Hamiltonian $\bar{\mathscr{K}}$ [see Eqs. (3–5)] reduce to

$$\bar{\mathscr{K}_0} = \mathscr{K}_0, \tag{37}$$

$$\bar{\mathscr{K}_1} = (\mathscr{K}_0, \bar{\mathscr{W}_1}) + \mathscr{K}_1, \tag{38}$$

$$\bar{\mathscr{K}_2} = \mathscr{K}_2 + (\bar{\mathscr{K}_1} + \mathscr{K}_1, \bar{\mathscr{W}_1}) + (\mathscr{K}_0, \bar{\mathscr{W}_2}), \tag{39}$$

where $\bar{\mathscr{K}}$ is chosen to consist of only the momenta and the long-period terms dependent on g. $\bar{\mathscr{W}}$ is the short-period generating function for the zonal harmonics. To obtain $\bar{\mathscr{K}_1}$, the average of $\mathscr{K}_1$ is computed with respect to l over the range $(0, 2\pi)$. The powers of the cosine function introduced by the inverse powers of r in $\mathscr{K}_1$ can be converted into a form suitable for integration using the following formulae (Gradshteyn and Ryzhik 2014):

$$(1 + e\cos f)^{n+1} = \sum_{s=0}^{n+1} \binom{n+1}{s} e^s \cos^s f, \tag{40}$$

$$\cos^s f = \frac{1}{2^{s-1}} \left\{ \sum_{k=0}^{\lfloor \frac{s-1}{2} \rfloor} \binom{s}{k} \cos((s-2k)f) + \frac{\lambda_s}{2} \binom{s}{\frac{s}{2}} \right\}, \tag{41}$$

where λ_s is defined as

$$\lambda_s = \begin{cases} 1 & \text{if } s \text{ is even} \\ 0 & \text{if } s \text{ is odd,} \end{cases} \tag{42}$$

and $\lfloor\ \rfloor$ represents the floor function. $\mathscr{K}$ is a function of the true anomaly, and in order to average it with respect to the mean anomaly, the following Keplerian relation between the true and mean anomalies is also needed:

$$\mathrm{d}l = \frac{r^2}{a^2\eta}\mathrm{d}f. \tag{43}$$

By using the preceding relationship and the trigonometric identities, the following integral is obtained, which must be evaluated to obtain $\bar{\mathscr{K}_1}$:

$$\begin{aligned} \bar{\mathscr{K}_1} &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n+2}} \sum_{p=0}^{p=n} F_{n0p}(i) A (1 + e\cos f)^{n+1} \frac{r^2}{a^2\eta} \mathrm{d}f, \\ &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n-1}} \sum_{p=0}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \\ &\quad \times \left\{ \sum_{k=0}^{\lfloor \frac{s-1}{2} \rfloor} \binom{s}{k} B + \lambda_s \binom{s}{\frac{s}{2}} A \right\} \mathrm{d}f, \end{aligned} \tag{44}$$

where

$$B = \begin{cases} \cos((n-2p-(s-2k))f + (n-2p)g) & \\ +\cos((n-2p+s-2k)f + (n-2p)g) & \text{if } n \text{ is even} \\ \sin((n-2p+s-2k)f + (n-2p)g) & \\ +\sin((n-2p-(s-2k))f + (n-2p)g) & \text{if } n \text{ is odd.} \end{cases}$$

The preceding integration can be performed by retaining only those terms in $\bar{\mathscr{K}_1}$ from $\mathscr{K}_1$ that do not depend on f. These terms are obtained when any of the three conditions: $n-2p=0$, $0 \le k = (n-2p+s)/2 \le \lfloor (s-1)/2 \rfloor$, and $0 \le k = (s-(n-2p))/2 \le \lfloor (s-1)/2 \rfloor$, is true. Therefore, the new Hamiltonian may be chosen as

$$\bar{\mathscr{K}_1} = \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n-1}} \left(F_{n0\frac{n}{2}}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \lambda_n \lambda_s \binom{s}{\frac{s}{2}} \right.$$
$$\left. + \sum_{p=0}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \left(\delta_{k'} \binom{s}{k'} C + \delta_{k''} \binom{s}{k''} C \right) \right), \tag{45}$$

where

$$\lambda_n = \begin{cases} 1 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd,} \end{cases}$$

$$\delta_{k'(k'')} \equiv \begin{cases} 1 & \text{if } k'(k'') \le \lfloor (s-1)/2 \rfloor \\ 0 & \text{otherwise,} \end{cases}$$

$$k' = \frac{n-2p+s}{2},$$
$$k'' = \frac{s-(n-2p)}{2},$$

$$C = \begin{cases} \cos((n-2p)g) & \text{if } n \text{ is even} \\ \sin((n-2p)g) & \text{if } n \text{ is odd.} \end{cases}$$

The above result provides a generalized expression for the short-period averaged Hamiltonian valid for an arbitrary zonal or tesseral harmonic, assuming no resonance effects due to any of the tesseral harmonics. Next, the homological equation given in Eq. (38) is solved to obtain the short-period generating function as shown below:

$$\bar{\mathscr{W}_1} = -\frac{1}{\frac{\partial \mathscr{K}_0}{\partial L}} \int \left(\bar{\mathscr{K}_1} - \mathscr{K}_1 \right) \mathrm{d}l,$$
$$= -\frac{1}{\frac{\partial \mathscr{K}_0}{\partial L}} \left\{ \bar{\mathscr{K}_1}(l-f) - \int_p \mathscr{K}_1 \frac{r^2}{a^2\eta} \mathrm{d}f \right\}, \tag{46}$$

where the subscript p indicates that the integration is performed only for the periodic terms of the integrand. Using the definition of $\bar{\mathscr{K}_1}$ and evaluating the preceding equation, $\bar{\mathscr{W}_1}$ is obtained as

$$\bar{\mathscr{W}_1} = \frac{-L^3}{\mu^2} \bar{\mathscr{K}_1}(l-f)$$
$$+ \sum_{n=2}^{\infty} \frac{\mu^n R_e^n J_n}{G^{2n-1}} \left(\sum_{\substack{p=0 \\ p \ne n/2}}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \lambda_s \binom{s}{\frac{s}{2}} D \right.$$
$$\left. + \sum_{p=0}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \left(\sum_{\substack{k=0 \\ k \ne k'}}^{\lfloor \frac{s-1}{2} \rfloor} \binom{s}{k} E' + \sum_{\substack{k=0 \\ k \ne k''}}^{\lfloor \frac{s-1}{2} \rfloor} \binom{s}{k} E'' \right) \right), \tag{47}$$

where

$$D = \int A \, \mathrm{d}f,$$
$$= \begin{cases} \frac{1}{n-2p} \sin((n-2p)(f+g)) & \text{if } n \text{ is even} \\ \frac{-1}{n-2p} \cos((n-2p)(f+g)) & \text{if } n \text{ is odd,} \end{cases}$$

$$E' = \begin{cases} \frac{1}{n-2p+(s-2k)} \sin((n-2p+(s-2k))f + (n-2p)g) & \text{if } n \text{ is even} \\ \frac{-1}{n-2p+(s-2k)} \cos((n-2p+(s-2k))f + (n-2p)g) & \text{if } n \text{ is odd,} \end{cases}$$

$$E'' = \begin{cases} \frac{1}{n-2p-(s-2k)} \sin((n-2p-(s-2k))f + (n-2p)g) & \text{if } n \text{ is even} \\ \frac{-1}{n-2p-(s-2k)} \cos((n-2p-(s-2k))f + (n-2p)g) & \text{if } n \text{ is odd.} \end{cases}$$

It is noted that the preceding results provide new closed-form generalized expressions for the first-order short-period averaged Hamiltonian in Eq. (45) and the first-order short-period generating function in Eq. (47) valid for an arbitrary zonal harmonic. These expressions are derived differently from the previous work done by De Saedeleer (2005) and arguably provide more compact representations.

The Hamiltonian, $\bar{\mathscr{K}}$, obtained after the short-period averaging is a function of the elements, which exhibit secular and long-period variations, and are equivalent to the single-primed elements of Brouwer (1959). In this work, primes or overbars to indicate mean elements are omitted for notational convenience. The osculating elements can be recovered from these elements by adding to them the short-period variations due to the zonal and tesseral harmonics. Considering Eqs. (6–7), the first-order short-period variations due to the zonal harmonics are obtained by evaluating the Poisson bracket of these elements with $\bar{\mathscr{W}}_1$. The results for the Delaunay elements are as given below:

$$\Delta\bar{l} = (\bar{l}, \bar{\mathscr{W}}_1) = \left(\frac{2L}{\mu}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial a} + \left(\frac{\eta^2}{eL}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial e} + \left(\frac{-G}{L^2}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial \eta}, \tag{48}$$

$$\Delta\bar{g} = (\bar{g}, \bar{\mathscr{W}}_1) = \left(\frac{-\eta^2}{eG}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial e} + \left(\frac{1}{L}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial \eta} + \left(\frac{\cos i}{G\sin i}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial i}, \tag{49}$$

$$\Delta\bar{h} = (\bar{h}, \bar{\mathscr{W}}_1) = \left(\frac{-1}{G\sin i}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial i}, \tag{50}$$

$$\Delta\bar{L} = (\bar{L}, \bar{\mathscr{W}}_1) = -\frac{\partial\bar{\mathscr{W}}_1}{\partial l} - \left(\frac{\mathrm{d}f}{\mathrm{d}l}\right)\frac{\partial\bar{\mathscr{W}}_1}{\partial f}, \tag{51}$$

$$\Delta\bar{G} = (\bar{G}, \bar{\mathscr{W}}_1) = -\frac{\partial\bar{\mathscr{W}}_1}{\partial g}, \tag{52}$$

$$\Delta\bar{H} = (\bar{H}, \bar{\mathscr{W}}_1) = -\frac{\partial\bar{\mathscr{W}}_1}{\partial h}, \tag{53}$$

where Δ and the overbar on the left hand side of the above equations indicate that these are short-period variations that must be added to the corresponding elements, whose time variations are determined by $\bar{\mathscr{K}}$, in order to recover the tesseral short-period averaged elements.

The application of the two short-period transformations sequentially, first for the sectorial and tesseral harmonics and then for the zonal harmonics, provides the benefit of including

some of the coupling effects indirectly, which would not be possible in the case of a single combined transformation. This is because when computing the osculating elements from the mean elements, the zonal short-period transformation is applied first and then the resulting elements are transformed into the osculating elements using the tesseral short-period transformation.

4 Long-period averaging

The long-period terms present in the short-period averaged Hamiltonian, $\bar{\mathscr{K}}$, are separated in this section in order to obtain a completely secular Hamiltonian, $\bar{\bar{\mathscr{K}}}$. The conventional way of performing the long-period transformation follows the same approach as used in case of the short-period transformations. The original homological equations given in Eqs. (3–5) simplify for this case to

$$\bar{\bar{\mathscr{K}}}_0 = \bar{\mathscr{K}}_0, \tag{54}$$

$$\bar{\bar{\mathscr{K}}}_1 = (\bar{\mathscr{K}}_0, \bar{\bar{\mathscr{W}}}_1) + \bar{\mathscr{K}}_1, \tag{55}$$

$$\bar{\bar{\mathscr{K}}}_2 = \bar{\mathscr{K}}_2 + (\bar{\bar{\mathscr{K}}}_1 + \bar{\mathscr{K}}_1, \bar{\bar{\mathscr{W}}}_1) + (\bar{\mathscr{K}}_0, \bar{\bar{\mathscr{W}}}_2). \tag{56}$$

The unperturbed Keplerian Hamiltonian, $\bar{\mathscr{K}}_0$, is degenerate in the angular momentum (G), and the long-period generating function, $\bar{\bar{\mathscr{W}}}$, is not a function of l; therefore, the preceding equations further simplify to

$$\bar{\bar{\mathscr{K}}}_0 = \bar{\mathscr{K}}_0, \tag{57}$$

$$\bar{\bar{\mathscr{K}}}_1 = \bar{\mathscr{K}}_1, \tag{58}$$

$$\bar{\bar{\mathscr{K}}}_2 = \bar{\mathscr{K}}_2 + 2(\bar{\mathscr{K}}_1, \bar{\bar{\mathscr{W}}}_1). \tag{59}$$

If the long-period terms are present in $\bar{\mathscr{K}}_1$, then these cannot be eliminated from $\bar{\bar{\mathscr{K}}}_1$ as evident in the preceding equations. The second zonal harmonic does not produce any long-period term when averaged with respect to the mean anomaly. Therefore, if J_2 is the only first-order perturbation in $\mathscr{H}_1$, a completely secular Hamiltonian can still be obtained (Brouwer 1959). However, this is not possible if an arbitrary spherical harmonic other than J_2 is dominant and needs to be included in $\mathscr{H}_1$.

In order to obtain a secular Hamiltonian in these cases, the degeneracy of the zeroth-order Hamiltonian with respect to the angular momentum must be addressed. This degeneracy may be removed by reordering the terms in $\bar{\mathscr{K}}$ such that the zeroth-order term, $\bar{\mathscr{K}}_0$, includes the Keplerian part and the secular part of $\bar{\mathscr{K}}_1$ from Eq. (45). That is, the short-period averaged Hamiltonian, $\bar{\mathscr{K}}$, may be rewritten as

$$\bar{\mathscr{K}} = \bar{\mathscr{K}}_0 + \epsilon \bar{\mathscr{K}}_1 + \mathscr{O}(\epsilon^2), \tag{60}$$

where

$$\bar{\mathscr{K}}_0 = -\frac{\mu^2}{2L^2} + \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n-1}} F_{n0\frac{n}{2}}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \lambda_n \lambda_s \binom{s}{\frac{s}{2}}, \tag{61}$$

$$\bar{\mathscr{K}}_1 = \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n-1}} \sum_{p=0}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \left(\delta_{k'} \binom{s}{k'} C + \delta_{k''} \binom{s}{k''} C \right). \tag{62}$$

As before, ϵ is a formal parameter, which is introduced only to separate the terms according to their respective order. $\bar{\mathscr{K}}_0$ in the preceding equation is a non-Keplerian Hamiltonian, and

therefore, the zeroth-order solution obtained is not a Keplerian ellipse. Only the even zonal harmonics contribute to $\bar{\mathscr{K}}_0$ due to the presence of λ_n term. It is also noted that J_2 has no long-period contributions to $\bar{\mathscr{K}}_1$. As a result, $\bar{\mathscr{K}}_1 = 0$ if J_2 is the only zonal harmonic considered and the first-order secular Hamiltonian $\bar{\mathscr{K}}$ for the main problem of artificial satellite is obtained. However, in the more general case, $\bar{\mathscr{K}}_1$ is not zero and the long-period terms must be separated from it. Considering the new form of $\bar{\mathscr{K}}$, the following homological equations are obtained up to second order:

$$\bar{\bar{\mathscr{K}}}_0 = \bar{\mathscr{K}}_0, \tag{63}$$

$$\begin{aligned}\bar{\bar{\mathscr{K}}}_1 &= \bar{\mathscr{K}}_1 + (\bar{\mathscr{K}}_0, \bar{\bar{\mathscr{W}}}_1),\\ &= \bar{\mathscr{K}}_1 - \frac{\partial \bar{\mathscr{K}}_0}{\partial G}\frac{\partial \bar{\bar{\mathscr{W}}}_1}{\partial g},\end{aligned} \tag{64}$$

$$\begin{aligned}\bar{\bar{\mathscr{K}}}_2 &= \bar{\mathscr{K}}_2 + (\bar{\bar{\mathscr{K}}}_1 + \bar{\mathscr{K}}_1, \bar{\bar{\mathscr{W}}}_1) + (\bar{\mathscr{K}}_0, \bar{\bar{\mathscr{W}}}_2),\\ &= \bar{\mathscr{K}}_2 + (\bar{\bar{\mathscr{K}}}_1 + \bar{\mathscr{K}}_1, \bar{\bar{\mathscr{W}}}_1) - \frac{\partial \bar{\mathscr{K}}_0}{\partial G}\frac{\partial \bar{\bar{\mathscr{W}}}_2}{\partial g}.\end{aligned} \tag{65}$$

It is noted from Eqs. (61) and (62) that $\bar{\mathscr{K}}_1$ is purely periodic and all the secular terms are in $\bar{\mathscr{K}}_0$. Therefore, $\bar{\bar{\mathscr{K}}}_1$ may be chosen to be equal to zero. The first-order long-period generating function for an arbitrary zonal harmonic is then computed from Eq. (64) as follows:

$$\begin{aligned}\bar{\bar{\mathscr{W}}}_1 &= \frac{1}{\frac{\partial \bar{\mathscr{K}}_0}{\partial G}} \int \bar{\mathscr{K}}_1 \mathrm{d}g,\\ &= \frac{1}{\frac{\partial \bar{\mathscr{K}}_0}{\partial G}} \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1}\eta^{2n-1}} \sum_{\substack{p=0\\ p\neq n/2}}^{p=n} F_{n0p}(i) \sum_{s=0}^{n-1} \binom{n-1}{s} \frac{e^s}{2^s} \left(\delta_{k'}\binom{s}{k'}P + \delta_{k''}\binom{s}{k''}P\right),\end{aligned} \tag{66}$$

where

$$\begin{aligned}P &= \int C\,\mathrm{d}g,\\ &= \begin{cases} \frac{1}{(n-2p)}\sin((n-2p)g) & \text{if } n \text{ is even}\\ \frac{-1}{(n-2p)}\cos((n-2p)g) & \text{if } n \text{ is odd.}\end{cases}\end{aligned}$$

It is noted that the critical inclination value is determined by equating the derivative of $\bar{\mathscr{K}}_0$ with respect to G to zero. Carrying out the differentiation of Eq. (61) with respect to G yields

$$\begin{aligned}\frac{\partial \bar{\mathscr{K}}_0}{\partial G} &= \frac{\partial \bar{\mathscr{K}}_0}{\partial e}\left(\frac{-\eta^2}{eG}\right) + \frac{\partial \bar{\mathscr{K}}_0}{\partial \eta}\left(\frac{1}{L}\right) + \frac{\partial \bar{\mathscr{K}}_0}{\partial i}\left(\frac{\cos i}{G\sin i}\right),\\ &= \sum_{n=2}^{\infty} \frac{\mu R_e^n J_n}{a^{n+1} G\,\eta^{2n-1}} \sum_{s=0}^{n-1} \frac{1}{2^s}\binom{n-1}{s}\lambda_n \lambda_s \binom{s}{\frac{s}{2}}\\ &\quad \times \left((1-2n)e^s F_{n0\frac{n}{2}}(i) - s\,e^{s-2}\,\eta^2 F_{n0\frac{n}{2}}(i) + e^s \frac{\cos i}{\sin i} F'_{n0\frac{n}{2}}(i)\right),\end{aligned} \tag{67}$$

where

$$F_{n0\frac{n}{2}}(i) = (-1)^{E\left(\frac{n-1}{2}\right)} \frac{(n)!}{2^n((n/2)!)^2} \sum_{j=0}^{j=n} (-1)^j \times \binom{n}{j}\binom{n}{n-j} \cos^{2n-2j}\left(\frac{i}{2}\right) \sin^{2j}\left(\frac{i}{2}\right).$$

A single prime on the term $F_{n0\frac{n}{2}}(i)$ represents its derivative with respect to i. The preceding equations are completely free of singularities because of cancellations of the singular terms. The critical inclination value is found by equating Eq. (67) to zero. It is concluded from the preceding results that the critical inclination value in the presence of all the zonal harmonics depends on the eccentricity value in the general case. It is noted that $\bar{\bar{\mathscr{W}}}_1$, in fact, has the magnitude of a zeroth-order term because of the presence of the critical inclination term [see Eq. (67)] in the denominator, which is a first-order term.

The first-order long-period variations are derived by evaluating the Poisson brackets $(\bar{\bar{\mathbf{X}}}, \bar{\bar{\mathscr{W}}}_1)$, where $\bar{\bar{\mathbf{X}}}$ are the mean elements. The results in the case of the Delaunay elements are

$$\Delta\bar{\bar{l}} = \left(\frac{2L}{\mu}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial a} + \left(\frac{\eta^2}{eL}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial e} + \left(\frac{-G}{L^2}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial \eta}, \tag{68}$$

$$\Delta\bar{\bar{g}} = \left(\frac{-\eta^2}{eG}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial e} + \left(\frac{1}{L}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial \eta} + \left(\frac{\cos i}{G\sin i}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial i}, \tag{69}$$

$$\Delta\bar{\bar{h}} = \left(\frac{-1}{G\sin i}\right)\frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial i}, \tag{70}$$

$$\Delta\bar{\bar{L}} = \frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial l} = 0, \tag{71}$$

$$\Delta\bar{\bar{G}} = \frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial g} = \frac{-1}{\frac{\partial\bar{\mathscr{K}}_0}{\partial G}}\bar{\mathscr{K}}_1, \tag{72}$$

$$\Delta\bar{\bar{H}} = \frac{\partial\bar{\bar{\mathscr{W}}}_1}{\partial h} = 0. \tag{73}$$

The elements used in the right hand side of the preceding equations are the mean elements, which are similar to the double-primed elements of Brouwer (1959). The secular variations of these mean elements can be computed using Hamilton's equations with $\bar{\bar{\mathscr{K}}}$ as the secular Hamiltonian. The mean action elements are constants in time, and the constant secular rates of the angle elements for an arbitrary zonal harmonic of degree n are obtained from

$$\dot{\bar{\bar{l}}} = \frac{2L}{\mu}\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial a} + \frac{\eta^2}{eL}\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial e} - \frac{G}{L^2}\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial \eta}, \tag{74}$$

$$\dot{\bar{\bar{g}}} = \left(\frac{-\eta^2}{eG}\right)\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial e} + \left(\frac{1}{L}\right)\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial \eta} + \left(\frac{\cos i}{G\sin i}\right)\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial i}, \tag{75}$$

$$\dot{\bar{\bar{h}}} = \left(\frac{-1}{G\sin i}\right)\frac{\partial\bar{\bar{\mathscr{K}}}}{\partial i}, \tag{76}$$

where the double overbars on the right hand side of the equations are omitted again for notational convenience. By virtue of Eq. (63), the expression for $\dot{\bar{\bar{g}}}$ is given by Eq. (67).

5 Applications

The analytic orbit theory developed in the previous sections is applied to the motion of two artificial satellites: the first is a lunar orbiter, and the second orbits the asteroid 433 Eros. The unnormalized spherical harmonic coefficients of the Moon[1] and 433 Eros[2] are given in Tables 1 and 2. Inspection of the Moon's coefficients shows that C_{20} has the largest magnitude followed by C_{31}, C_{22}, and C_{60}. The latter three are of the same order of magnitude and only one order of magnitude smaller than the former. In fact, the ratio of the magnitudes of C_{31} to C_{20} is 0.14. For 433 Eros, C_{22}, S_{22}, C_{40}, and C_{60} are one order of magnitude smaller than the oblateness coefficient. The ratios of C_{40} to C_{20} and C_{22} to C_{20} are 0.33 and 0.45, respectively. In the case of Earth, C_{20} is at least three orders of magnitude larger than any other coefficient and the largest ratio of the magnitudes of any coefficient to that of C_{20} is only 0.002. Therefore, an orbit theory for 433 Eros must include C_{22} and C_{40} along with C_{20} as the first-order perturbations. It is noted that for a complete theory for a lunar orbiter, the third-body perturbation due to Earth has significant effects and needs to be included (De Saedeleer 2006). However, in the present work, the analytic theory for the lunar orbiter only considers the gravitational perturbations and the third-body effects are ignored, which can be incorporated using the conventional approaches as shown by De Saedeleer (2006).

The analytical expressions as derived for the first-order secular Hamiltonian and generating functions in the previous sections are all expressed in a generalized form valid for any arbitrary spherical harmonic. As a result, a first-order orbit theory can be easily built using these generalized expressions with spherical harmonics included up to any degree and order. In the present work, these generalized expressions are implemented in a symbolic algebra software package and their partial derivatives are computed symbolically for zonal harmonics from J_2 to J_6 in order to compute the expressions for the secular rates, short-period, and long-period variations of the Delaunay elements. The expressions for the short-period variations due to the sectorial and tesseral harmonics are derived by hand. For demonstration, these analytical expressions are used to construct two orbit theories: the first for the lunar orbiter and the second for the 433 Eros orbiter, with all the spherical harmonics up to 6×6 considered dominant perturbations. As discussed earlier, the long-period or resonant effects of the sectorial and tesseral harmonics are not considered in the present work.

If all the spherical harmonics up to 6×6 are included in the $\mathscr{H}_1$ term of the original Hamiltonian given in Eq. (14), then the secular Hamiltonian, $\bar{\bar{\mathscr{K}}}$, up to first order is

$$\bar{\bar{\mathscr{K}}} = \bar{\mathscr{K}}_0 + \mathscr{O}(\epsilon^2). \tag{77}$$

By evaluating the expression given in Eq. (61) from $n = 2$ to $n = 6$, $\bar{\bar{\mathscr{K}}}$ is obtained as

$$\bar{\bar{\mathscr{K}}} = -\frac{\mu}{2\,L^2} + \bar{\bar{\mathscr{K}}}|_{n=2} + \bar{\bar{\mathscr{K}}}|_{n=4} + \bar{\bar{\mathscr{K}}}|_{n=6} + \mathscr{O}(\epsilon^2), \tag{78}$$

where

$$\bar{\bar{\mathscr{K}}}|_{n=2} = \frac{\mu R_e^2 J_2}{a^3 \eta^3}\Big(F_{201}(i) + F_{200}(i) \times 0 + F_{201}(i) \times 0 + F_{202}(i) \times 0\Big),$$

[1] Data retrieved on March 30, 2019, from http://pds-geosciences.wustl.edu/grail/grail-l-lgrs-5-rdr-v1/grail_1001/shadr/gggrx_1200a_sha.tab.

[2] Data retrieved on March 30, 2019, from https://sbn.psi.edu/pds/resource/nearbrowse.html.

Table 1 The Moon's gravity model GRGM1200A unnormalized coefficients as measured by GRAIL mission

n	m	C_{nm}	S_{nm}
2	0	**−0.000203223560871**	0
2	1	$1.89313020989899\times10^{-11}$	$1.5146933495568\times10^{-09}$
2	2	$\mathbf{2.2381387532075\times10^{-05}}$	$5.86059183759659\times10^{-10}$
3	0	$-8.459342178393\times10^{-06}$	0
3	1	$\mathbf{2.84807022777237\times10^{-05}}$	$5.89158264904969\times10^{-06}$
3	2	$4.84049859737469\times10^{-06}$	$1.66616668871111\times10^{-06}$
3	3	$1.71169602649627\times10^{-06}$	$-2.4740346309803\times10^{-07}$
4	0	$9.70434253277103\times10^{-06}$	0
4	1	$-5.70489105972404\times10^{-06}$	$1.57891241626164\times10^{-06}$
4	2	$-1.59122222069953\times10^{-06}$	$-1.51539381224694\times10^{-06}$
4	3	$-8.0670152605846\times10^{-08}$	$-8.03491032736319\times10^{-07}$
4	4	$-1.2692023976215\times10^{-07}$	$8.29606171159122\times10^{-08}$
5	0	$-7.42211918696389\times10^{-07}$	0
5	1	$-8.66279054424634\times10^{-07}$	$-3.52722195988995\times10^{-06}$
5	2	$7.11995662349507\times10^{-07}$	$1.71081805759989\times10^{-07}$
5	3	$1.53985511643403\times10^{-08}$	$2.87362117301303\times10^{-07}$
5	4	$2.14453844519568\times10^{-08}$	$5.26955848745972\times10^{-10}$
5	5	$7.6589150664677\times10^{-09}$	$-6.7820126340736\times10^{-09}$
6	0	$\mathbf{1.37675387844709\times10^{-05}}$	0
6	1	$1.20242758346682\times10^{-06}$	$-2.04535071385399\times10^{-06}$
6	2	$-5.47039455997804\times10^{-07}$	$-2.69668097631117\times10^{-07}$
6	3	$-6.87848776295172\times10^{-08}$	$-7.10638407618996\times10^{-08}$
6	4	$1.2917397302938\times10^{-09}$	$-1.53616610816221\times10^{-08}$
6	5	$1.1738100266618\times10^{-09}$	$-8.34649273663825\times10^{-09}$
6	6	$-1.09132983765608\times10^{-09}$	$1.68440977528971\times10^{-09}$

The reference radius and gravitational constant are 1738 km and 4902.8001224453001 km^3/s^2, respectively. The value of the Moon's sidereal rotation period is used as 27.3220 days. The values with the largest and the second largest order of magnitude are highlighted in bold

$$
\begin{aligned}
&= J_2 \frac{\mu^4 R_e^2}{4L^3G^3}\left(1-3\frac{H^2}{G^2}\right),\\
\bar{\bar{\mathscr{K}}}|_{n=4} &= J_4 \frac{3\,\mu^6 R_e^4}{128\,L^3G^7}\left(5-3\frac{G^2}{L^2}\right)\left(3-30\frac{H^2}{G^2}+35\frac{H^4}{G^4}\right),\\
\bar{\bar{\mathscr{K}}}|_{n=6} &= J_6 \frac{5\,\mu^8 R_e^6}{2048\,L^3G^{11}}\left(63-70\frac{G^2}{L^2}+15\frac{G^4}{L^4}\right)\left(5-105\frac{H^2}{G^2}+315\frac{H^4}{G^4}-231\frac{H^6}{G^6}\right).
\end{aligned}
$$

It is noted that only the even harmonics contribute to the first-order secular Hamiltonian. Using the preceding results, the secular variations of the mean Delaunay elements are computed up to first order in a straightforward manner by using Eqs. (74–76).

Table 2 433 Eros gravity model unnormalized coefficients from NEAR mission

n	m	C_{nm}	S_{nm}
2	0	**−0.11730823892116**	0
2	1	$-2.11453654036999\times10^{-06}$	$-1.80744136559608\times10^{-07}$
2	2	**0.0531885762344478**	**−0.0181446403126972**
3	0	−0.00374232360832355	0
3	1	0.00438547916269853	0.00363836234929177
3	2	0.000606590143163718	−0.000240394551651026
3	3	−0.00145249757637031	−0.00168328195232911
4	0	**0.0387944459886**	0
4	1	$-9.52012546349172\times10^{-05}$	0.00012991262353829
4	2	−0.00390614283829429	0.00103509812252799
4	3	$-1.79404526644834\times10^{-05}$	$-7.12399065781978\times10^{-06}$
4	4	0.000368807714106221	−0.00019238442311687
5	0	0.00218543636869044	0
5	1	−0.0023678671891483	−0.00104272783346457
5	2	−0.000126781332202482	$6.17062269039374\times10^{-05}$
5	3	0.0001511694114223	0.00011692462186204
5	4	$3.8690753546703\times10^{-06}$	$-5.43530823252772\times10^{-06}$
5	5	$-2.51306515430204\times10^{-05}$	$-1.43782460033982\times10^{-05}$
6	0	**−0.0179497621895508**	0
6	1	$-1.9165115025489\times10^{-05}$	$-9.74106192828274\times10^{-05}$
6	2	0.000813891437693812	−0.000148125756779844
6	3	$5.96640184914192\times10^{-06}$	$1.56395119542098\times10^{-06}$
6	4	$-2.13764251724757\times10^{-05}$	$6.69541971569532\times10^{-06}$
6	5	$-3.93777188512343\times10^{-07}$	$3.86798920130381\times10^{-08}$
6	6	$1.18484157149138\times10^{-06}$	$-3.73277755394728\times10^{-07}$

The reference radius and gravitational constant are 16 km and 0.000446275472004 km^3/s^2, respectively. The rotational angular speed is 1639.389232 degrees per day

Next, the long-period generating function $\bar{\bar{\mathscr{W}}}_1$ given in Eq. (66) is computed for all the zonal harmonics up to J_6. As discussed earlier, $\bar{\bar{\mathscr{W}}}_1$ has a singularity at the critical inclination and its value is computed by equating Eq. (67) to zero and solving for the value of the inclination in terms of a and e. Therefore, in general the value of the critical inclination depends on the semimajor axis and eccentricity of the orbit. For a circular orbit, the resulting equation to determine the value of the critical inclination up to first-order in the presence of zonal harmonics up to J_6 is

$$\begin{aligned}&\left(960\ (\cos(i))^2-192\right)J_2a^4-120\,{R_e}^2J_4\left(49\ (\cos(i))^4-36\ (\cos(i))^2+3\right)a^2\\&\quad+105\,{R_e}^4J_6\left(297\ (\cos(i))^6-375\ (\cos(i))^4+115\ (\cos(i))^2-5\right)=0.\end{aligned}\tag{79}$$

With the Moon as the central body, the values for the critical inclination for a circular orbit with semimajor axis equal to 1850 km are found to be 60.38° and 119.61° approximately. For 433 Eros, the critical inclinations for a orbit with semimajor axis equal to 50 km are

found as 62.78° and 117.21° approximately. If the oblateness term is the only perturbation considered, i.e., for $n = 2$, the preceding equation reduces to $1 - 5\cos^2 i = 0$ for any value of the semimajor axis and eccentricity, which gives the well-known values, 63.4° and 116.6°, for the critical inclination. The explicit expressions for $\bar{\bar{\mathscr{W}}}_1$ for zonal harmonics up to J_6 are

$$\bar{\bar{\mathscr{W}}}_1 = \bar{\bar{\mathscr{W}}}_1|_{n=2} + \bar{\bar{\mathscr{W}}}_1|_{n=3} + \bar{\bar{\mathscr{W}}}_1|_{n=4} + \bar{\bar{\mathscr{W}}}_1|_{n=5} + \bar{\bar{\mathscr{W}}}_1|_{n=6}, \tag{80}$$

where

$$\bar{\bar{\mathscr{W}}}_1|_{n=2} = 0, \tag{81}$$

$$\bar{\bar{\mathscr{W}}}_1|_{n=3} = 3/8\,\frac{e\mu^5 J_3 {R_e}^3 \cos(g)\sin(i)\left(5\,(\cos(i))^2 - 1\right)}{\dot{\bar{\bar{g}}}\,L^8\eta^5}, \tag{82}$$

$$\bar{\bar{\mathscr{W}}}_1|_{n=4} = \frac{15\,e^2\mu^6 J_4 {R_e}^4 \sin(2\,g)\,(\sin(i))^2\left(7\,(\cos(i))^2 - 1\right)}{128\,\dot{\bar{\bar{g}}}\,L^{10}\eta^7}, \tag{83}$$

$$\bar{\bar{\mathscr{W}}}_1|_{n=5} = \left(-\frac{15\,\sin(i)e\left(3\,e^2+4\right)\left(21\,ci^4 - 14\,ci^2 + 1\right)\cos(g)}{128\,\dot{\bar{\bar{g}}}\,L^{12}\eta^9} + \frac{35\,\sin(i)e^3\left(9\,ci^4 - 10\,ci^2 + 1\right)\cos(3\,g)}{768\,\dot{\bar{\bar{g}}}\,L^{12}\eta^9}\right) J_5\mu^7 {R_e}^5, \tag{84}$$

$$\bar{\bar{\mathscr{W}}}_1|_{n=6} = \left(\frac{-525\,e^2\sin^2(i)\,\sin(2\,g)\left(e^2+2\right)\left(33\,ci^4 - 18\,ci^2 + 1\right)}{4096\,\dot{\bar{\bar{g}}}\,L^{14}\eta^{11}} - \frac{315\,e^4\sin^4(i)\left(11\,ci^2 - 1\right)\sin(4\,g)}{16384\,\dot{\bar{\bar{g}}}\,L^{14}\eta^{11}}\right) J_6\mu^8 {R_e}^6, \tag{85}$$

and $ci = \cos(i)$. In the preceding equations, $\dot{\bar{\bar{g}}}$ is equal to the partial derivative of $\bar{\bar{\mathscr{K}}}$ given in Eq. (78) with respect to G. The long-period variations for the Delaunay elements are computed by substituting the expression for $\bar{\bar{\mathscr{W}}}_1$ in Eqs. (68–73). These long-period variations are added to the mean elements to recover the short-period averaged elements, which are equivalent to the single-primed elements of Brouwer (1959). It is noted, however, that the contribution of J_2 in the first-order long-period generating function is only through the critical inclination term $\dot{\bar{\bar{g}}}$. This is in contrast to Brouwer (1959), in which the first-order long-period contribution of J_2 is computed from the second-order terms.

The osculating elements are recovered by carrying out two short-period transformations, separately for the zonal and tesseral harmonics. The zonal short-period effects are computed using the generating function $\bar{\mathscr{W}}_1$ given in Eq. (47). The contribution of J_2 term to $\bar{\mathscr{W}}_1$ is

$$\begin{aligned}\bar{\mathscr{W}}_1|_{n=2} = {} & \frac{-L^3}{\mu^2}\bar{\mathscr{K}}_1(l - f) \\ & + \frac{\mu^2 R_e^2 J_2}{G^3}\Big[(F_{200}(i) + F_{202}(i))\sin(2f + 2g) + (F_{200}(i) + F_{202}(i))\,\frac{e}{2} \\ & \times (\sin(f + 2g) + \sin(3f + 2g)) + F_{201}(i)e\sin(f)\Big],\end{aligned}$$

$$= J_2 \frac{\mu^2 R_e^2}{4G^3} \left[\left(1 - 3\frac{H^2}{G^2}\right)(f - l + e\sin(f)) \right.$$
$$\left. -\frac{1}{2}\left(1 - \frac{H^2}{G^2}\right)(3e\sin(f + 2g) + 3\sin(2f + 2g) + e\sin(3f + 2g)) \right]. \quad (86)$$

Since the zonal short-period transformation is same as the one originally used by Brouwer (1959), the preceding expression matches exactly with his results. The expressions for the rest of the zonal harmonics can be similarly computed from Eq. (47) in a straightforward manner. Due to space limitations, these expressions are not provided here; however, the final results up to J_6 are same as the ones provided by Mahajan (2018). Substituting $\bar{\mathscr{W}}_1$ in Eqs. (48–53), the first-order short-period variations of the Delaunay elements due to the zonal harmonics can be computed. These expressions are derived in the present work using a symbolic algebra software package.

The final transformation involves computation of short-period effects due to the tesseral harmonics. For C_{22}/S_{22}, the expression for the tesseral short-period generating function can be evaluated using Eq. (20) and the result is

$$\begin{aligned}
\mathscr{W}_1 = &- \frac{\mu R_e^2}{\tilde{n}\, a^3 \eta^3} \left(\sum_{p=0}^{p=n} \frac{3(1+\cos i)^2}{4} \left[C_{22} \{\cos(\beta_0) I_1^{220} - \sin(\beta_0) I_2^{220}\} \right.\right. \\
&+ \quad S_{22} \{\cos(\beta_0) I_2^{220} + \sin(\beta_0) I_1^{220}\} \Big] \\
&+ \frac{3\sin^2 i}{2} \left[C_{22} \{\cos(\beta_1) I_1^{221} - \sin(\beta_1) I_2^{221}\} \right. \\
&+ S_{22} \{\cos(\beta_1) I_2^{221} + \sin(\beta_1) I_1^{221}\} \Big] \\
&+ \frac{3(1-\cos i)^2}{4} \left[C_{22} \{\cos(\beta_2) I_1^{222} - \sin(\beta_2) I_2^{222}\} \right. \\
&\left.\left. + S_{22} \{\cos(\beta_2) I_2^{222} + \sin(\beta_2) I_1^{222}\} \Big] \right), \qquad (87)
\end{aligned}$$

where

$$\beta_0 \equiv -2g + 2(h - \theta + \delta l), \quad (88)$$
$$\beta_1 \equiv 2(h - \theta + \delta l), \quad (89)$$
$$\beta_2 \equiv 2g + 2(h - \theta + \delta l), \quad (90)$$

and the expressions for the quadratures I_1^{nmp} and I_2^{nmp} for $n = 2, m = 2, p = 0, 1, 2$ are evaluated using Eqs. (22–23). By substituting $\mathscr{W}_1$ in Eqs. (24–29), the short-period variations of the Delaunay elements due to the sectorial and tesseral harmonics can be computed. In the present work, the generalized expressions for the tesseral short-period variations are derived by hand using the expression of $\mathscr{W}_1$ given in Eq. (20).

In order to verify the accuracy of the proposed theory, the analytic expressions for the secular rates, the long-period, and short-period variations of the Delaunay elements are implemented in a MATLAB simulation, which includes the contributions from all the spherical harmonics up to 6×6. The *ode45* numerical integrator of MATLAB, with a force model also consisting of 6×6 gravitational field, is used to generate the true trajectories. First, a lunar orbiter is simulated by choosing the initial osculating elements as given in Table 3. Its numerically propagated orbit for a time duration equal to 300 orbital periods is shown in Fig. 1. For

Table 3 Initial osculating classical elements of the lunar orbiter and 433 Eros orbiter

Orbiter	SMA(km)	Ecc	Inc(°)	RAAN(°)	AOP(°)	MA(°)
Lunar	1850	0.05	50	50	50	50
433 Eros	50	0.05	50	50	50	50

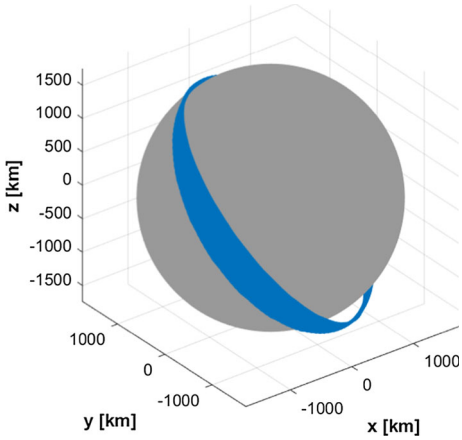

Fig. 1 Numerically propagated orbit of the lunar orbiter in a 6×6 gravity field. The Moon is drawn to scale

the analytical theory, the initial mean elements were computed first by using the analytical osculating-to-mean transformation. The resulting mean elements were then analytically propagated to the desired time and converted to osculating elements using the mean-to-osculating transformation. Figures 2 and 3 show the plots of the numerically propagated osculating elements, analytically propagated osculating elements, and the mean elements for a time duration equivalent to 20 orbit periods. The results show that the first-order orbit theory is able to reproduce the variations in the orbital elements quite well for the time duration considered. The mean elements are able to capture the secular dynamics; however, the plots show bias between the osculating and mean elements except for the semimajor axis and mean anomaly. This apparent bias is due to the long-period variations. Figure 4 shows the variation of the classical elements for a longer time duration of 240 days. The osculating elements in this figure are obtained by only carrying out the zonal long-period and short-period transformations (no tesseral short-period transformation) of the mean elements. The long-period variations of the eccentricity, inclination, and argument of the periapsis are clearly seen in these plots. Figure 5 shows the similar plots, in which the osculating elements also include the tesseral short-period effects. The variations of these osculating elements closely match that of the numerically propagated elements as seen in the figure. The position and velocity errors of the analytic orbit theory, plotted in Fig. 6, are less than 1 km and 0.02 km/s after 1.6 days of propagation, respectively.

Another simulation of an artificial satellite with 433 Eros as the central body is implemented using the initial conditions as given in Table 3. The true trajectory was again obtained by numerical propagation in MATLAB using a force model with 6×6 gravity model and is plotted in Fig. 7 for a duration of 30 orbit periods. The numerically and analytically propagated classical orbital elements are plotted in Figs. 8 and 9. The results show that the first-order analytic orbit theory captured the element variations in this case better than that for the lunar orbiter. This is due to the fact that the initial semimajor axis (=50 km) of the

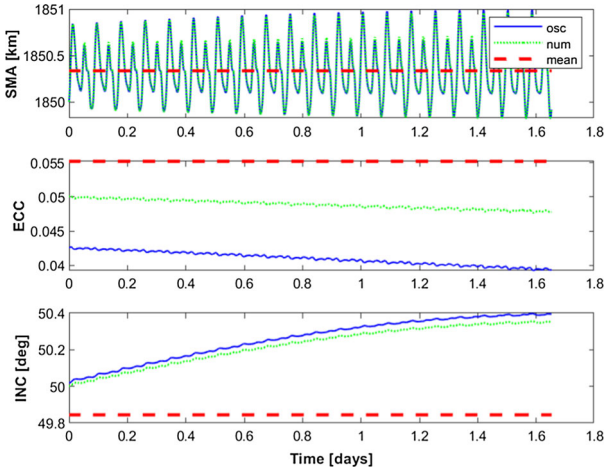

Fig. 2 First three osculating classical elements of the lunar orbiter propagated using numerical integration (num) and the analytic theory (osc). The mean elements (mean) are also shown

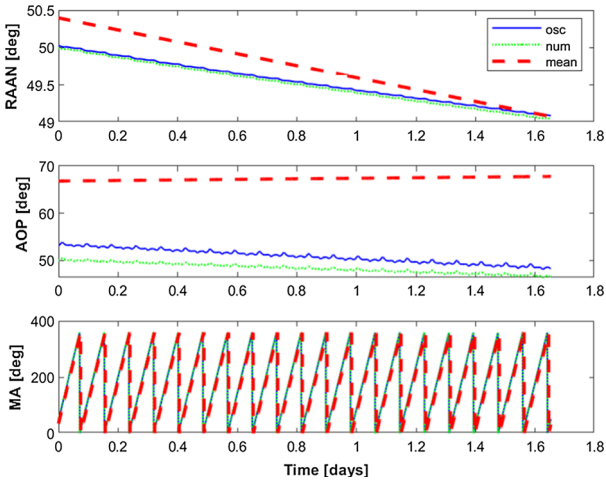

Fig. 3 Last three osculating classical elements of the lunar orbiter propagated using numerical integration (num) and the analytic theory (osc). The mean elements (mean) are also shown

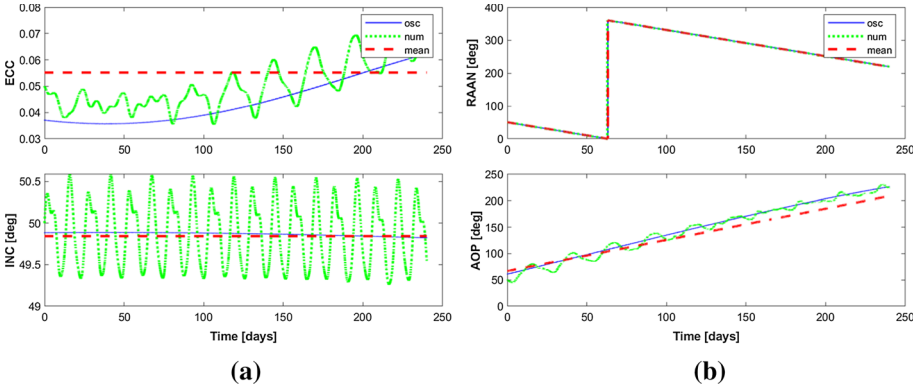

Fig. 4 The mean elements (mean), numerically propagated elements (num), and the elements obtained after the mean-to-zonal short-period transformation (osc) for the lunar orbiter

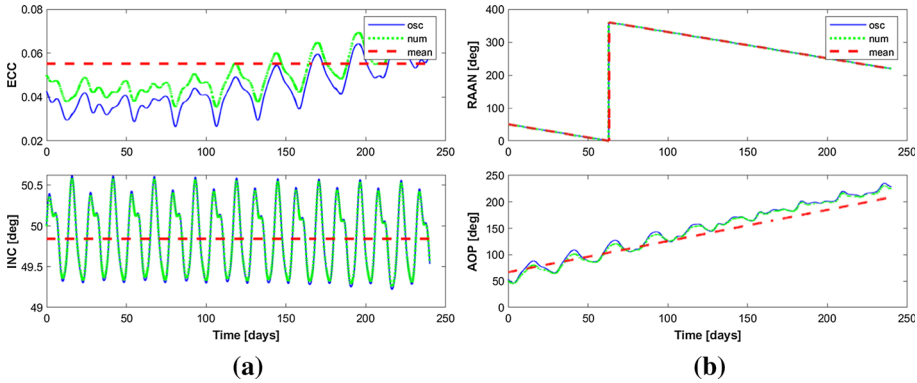

Fig. 5 The mean elements (mean), numerically propagated elements (num), and the elements obtained after the complete mean-to-osculating transformation (osc) for the lunar orbiter

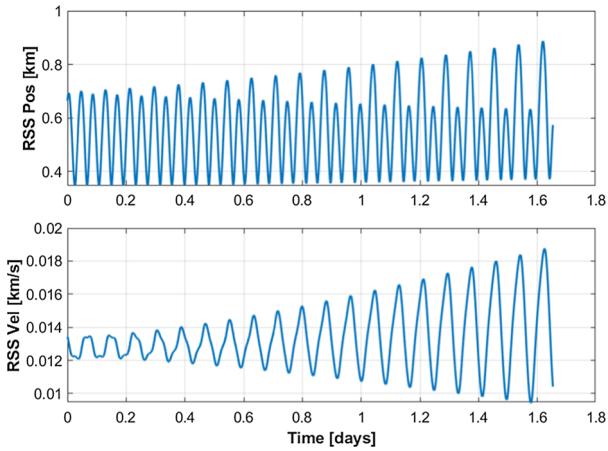

Fig. 6 Root-sum-square position and velocity errors of the analytic orbit theory for the lunar orbiter

chosen orbit in the case of 433 Eros orbiter is more than three times the reference radius (=16 km) of 433 Eros, whereas for the lunar orbiter the relative size of the orbit is much smaller. The position and velocity errors for the 433 Eros orbiter are shown in Fig. 10 for a duration of 10 orbit periods. The position errors remain below 0.1 km after nearly 12 days of propagation.

6 Conclusions

In general, the short-period averaging of the zonal spherical harmonics with degree greater than two results in secular and long-period terms. If these spherical harmonics are treated as second-order perturbations with the oblateness term as the first-order perturbation, then the second-order long-period terms can be removed by averaging with respect to the argument of perigee for constructing satellite theories of Earth. However, for bodies with more deformed shapes, it may be necessary to consider the higher spherical harmonics as the dominant perturbation. The conventional approach to separating these long-period terms from the Hamiltonian fails. A novel method to carry out the long-period transformations is developed,

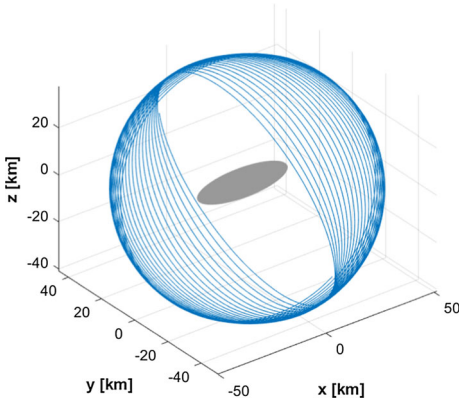

Fig. 7 Numerically propagated orbit of the satellite of 433 Eros asteroid in a 6 × 6 gravity field. 433 Eros is drawn to scale with its shape approximated as an ellipsoid with axes equal to 20.6 km, 5.7 km, and 5.3 km

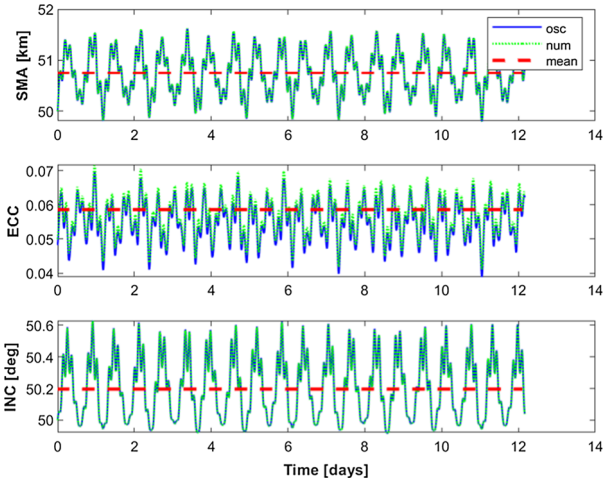

Fig. 8 First three classical orbital elements of the 433 Eros orbiter. The plots show numerically propagated (num), analytically propagated osculating (osc), and the mean (mean) elements

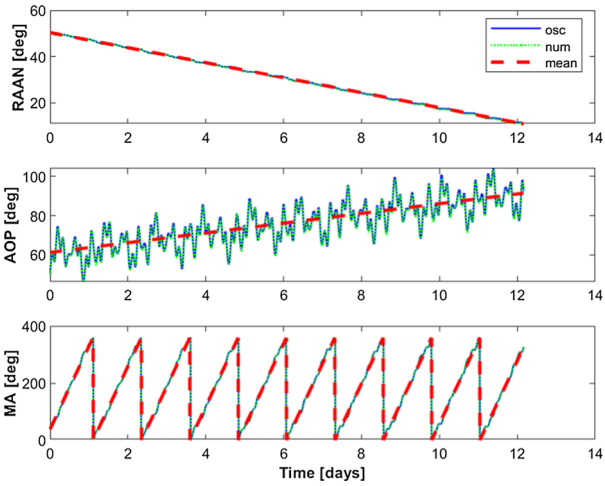

Fig. 9 Last three classical orbital elements of the 433 Eros orbiter. The plots show numerically propagated (num), analytically propagated osculating (osc), and the mean (mean) elements

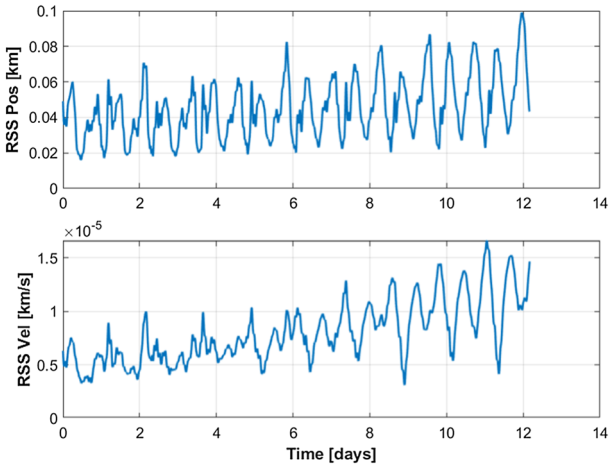

Fig. 10 Root-sum-square position and velocity errors of the analytic orbit theory for the 433 Eros orbiter

and it results in a completely secular Hamiltonian. As part of the method, the degeneracy of the Keplerian or zeroth-order Hamiltonian is eliminated by choosing a non-Keplerian Hamiltonian that consists of the two-body part as well as the first-order secular terms of all the zonal harmonics. This new approach results in a long-period term which is of zeroth order, not first order as in Brouwer's theory. The analytic expressions for the secular, short-period, and long-period effects are derived in a generalized form valid for any spherical harmonic. Results presented for the motion of a lunar orbiter and a 433 Eros orbiter in a 6×6 gravity field, show that the proposed first-order analytic theory is in agreement with the results obtained from the numerical integration.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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